

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		39172	
Total Threats Mitigated	42	Total Packets Analyzed		796961699	
High Priority	13	Malicious Packets		795025683	
Low Priority	29	True Negatives Passed		33704	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		16	
False Positives	0	UDP Flood		7	
FP Rate	0.00%	Uncertain		13	
IF Precision	100.00%				
IF Recall	95.56%	Action		Count	
IF F1-Score	97.73%	Quarantined		10	
IF Accuracy	99.40%	Time Ban		10	
		Blackhole		0	
		Released		5	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.56%	Share of actual attacks successfully flagged
F1-Score	97.73%	Balanced measure combining Precision and Recall
Accuracy	99.40%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.44%	Actual attacks that went undetected
True Positive Rate (TPR)	95.56%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4948	FN 230
Actual: Normal	FP 0	TN 33474

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	2328	0	0
Act: ICMP	0	919	0
Act: UDP	0	0	1701

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.56%	Share of actual attacks successfully flagged
F1-Score	97.73%	Balanced measure combining Precision and Recall
Accuracy	99.40%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.44%	Actual attacks that went undetected
True Positive Rate (TPR)	95.56%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4948	FN 230
Actual: Normal	FP 0	TN 33474

2d. Response Latency

Metric	Value	Description
Detection Time	1036.8 ms	Average time taken to detect an attack
Mitigation Response Time	16.3 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	3.13%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	36.70%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	41.96%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	4	3	2026-09-25 14:26:11	2026-09-25 14:28:51
10.0.0.17	4	3	2026-09-25 14:26:12	2026-09-25 14:28:52
10.0.0.18	3	3	2026-09-25 14:26:12	2026-09-25 14:27:23
10.0.0.19	3	3	2026-09-25 14:26:17	2026-09-25 14:27:28
10.0.0.20	3	2	2026-09-25 14:26:41	2026-09-25 14:28:52
10.0.0.21	3	2	2026-09-25 14:26:41	2026-09-25 14:29:11
10.0.0.22	2	2	2026-09-25 14:26:43	2026-09-25 14:27:24
10.0.0.23	2	1	2026-09-25 14:27:12	2026-09-25 14:28:52

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.24	2	1	2026-09-25 14:27:12	2026-09-25 14:28:55
10.0.0.25	2	1	2026-09-25 14:27:13	2026-09-25 14:29:01

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:26:11	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:12	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:12	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:17	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:21	10.0.0.16	DDoS	SYN Flood	91.7%	Critical	Time Ban (30s)
2026-09-25 14:26:22	10.0.0.16	DDoS	SYN Flood	91.7%	Critical	Time Ban
2026-09-25 14:26:22	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban (30s)
2026-09-25 14:26:23	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban (30s)
2026-09-25 14:26:23	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-09-25 14:26:24	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-09-25 14:26:28	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban (30s)
2026-09-25 14:26:28	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban
2026-09-25 14:26:41	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:41	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:43	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:26:52	10.0.0.16	DDoS	SYN Flood	93.6%	High	Released
2026-09-25 14:26:52	10.0.0.20	DDoS	UDP Flood	76.1%	High	Time Ban (30s)
2026-09-25 14:26:52	10.0.0.21	DDoS	UDP Flood	82.3%	High	Time Ban (30s)
2026-09-25 14:26:52	10.0.0.21	DDoS	UDP Flood	82.3%	High	Time Ban
2026-09-25 14:26:52	10.0.0.20	DDoS	UDP Flood	76.1%	High	Time Ban
2026-09-25 14:26:53	10.0.0.17	DDoS	SYN Flood	93.3%	High	Released
2026-09-25 14:26:54	10.0.0.22	DDoS	UDP Flood	78.6%	High	Time Ban (30s)
2026-09-25 14:26:54	10.0.0.22	DDoS	UDP Flood	78.6%	High	Time Ban
2026-09-25 14:27:02	10.0.0.16	DDoS	SYN Flood	93.6%	Critical	Time Ban (1m)
2026-09-25 14:27:04	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-25 14:27:12	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:27:12	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:27:13	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 14:27:22	10.0.0.23	DDoS	ICMP Flood	91.0%	High	Time Ban (30s)
2026-09-25 14:27:22	10.0.0.24	DDoS	ICMP Flood	91.3%	High	Time Ban (30s)
2026-09-25 14:27:22	10.0.0.24	DDoS	ICMP Flood	91.3%	High	Time Ban
2026-09-25 14:27:22	10.0.0.23	DDoS	ICMP Flood	91.0%	Critical	Time Ban
2026-09-25 14:27:23	10.0.0.18	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-25 14:27:23	10.0.0.25	DDoS	ICMP Flood	91.3%	High	Time Ban (30s)
2026-09-25 14:27:23	10.0.0.25	DDoS	ICMP Flood	91.3%	Critical	Time Ban
2026-09-25 14:27:28	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban (1m)
2026-09-25 14:28:51	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 14:28:52	10.0.0.20	DDoS	UDP Flood	78.2%	High	Time Ban (1m)
2026-09-25 14:28:52	10.0.0.17	DDoS	SYN Flood	90.1%	Critical	Time Ban (2m)
2026-09-25 14:28:55	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 14:29:01	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-25 14:29:11	10.0.0.21	DDoS	Uncertain	-	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 14:29:11	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:29:01	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-25 14:28:55	Attack Stopped
10.0.0.17	SYN Flood	0.6266	90.1%	Critical	Phase 2	1	11s	2026-09-25 14:28:52	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6448	93.9%	Critical	Phase 2	1	11s	2026-09-25 14:28:52	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6441	78.2%	High	Phase 2	1	11s	2026-09-25 14:28:52	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6492	93.1%	Critical	Phase 2	1	11s	2026-09-25 14:28:51	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6418	92.2%	Critical	Phase 2	3	1m 11s	2026-09-25 14:27:28	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6391	79.1%	High	Phase 2	2	41s	2026-09-25 14:27:24	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6456	91.3%	High	Phase 2	1	10s	2026-09-25 14:27:23	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6341	93.3%	Critical	Phase 2	3	1m 11s	2026-09-25 14:27:23	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6448	91.3%	High	Phase 2	1	10s	2026-09-25 14:27:22	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6451	91.0%	High	Phase 2	1	10s	2026-09-25 14:27:22	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6290	82.3%	High	Phase 2	2	41s	2026-09-25 14:27:22	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6128	82.5%	High	Phase 2	2	41s	2026-09-25 14:27:22	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6332	93.3%	Critical	Phase 2	3	11s	2026-09-25 14:27:04	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6324	93.6%	Critical	Phase 2	3	10s	2026-09-25 14:27:02	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6145	92.2%	Critical	Phase 2	2	41s	2026-09-25 14:26:58	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	78.6%	High	Phase 2	1	11s	2026-09-25 14:26:54	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6015	93.3%	High	Phase 2	1	41s	2026-09-25 14:26:53	Ban Expired
10.0.0.18	SYN Flood	0.6093	93.3%	Critical	Phase 2	2	41s	2026-09-25 14:26:53	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6297	82.3%	High	Phase 2	1	11s	2026-09-25 14:26:52	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	76.1%	High	Phase 2	1	11s	2026-09-25 14:26:52	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6030	93.6%	High	Phase 2	1	41s	2026-09-25 14:26:52	Ban Expired
10.0.0.19	SYN Flood	0.6465	92.2%	Critical	Phase 2	1	11s	2026-09-25 14:26:28	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6393	93.3%	Critical	Phase 2	1	11s	2026-09-25 14:26:23	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6434	93.3%	Critical	Phase 2	1	10s	2026-09-25 14:26:22	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6376	91.7%	Critical	Phase 2	1	10s	2026-09-25 14:26:21	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE